Package ‘dsClientHarmonization’

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Title What the Package Does (One Line, Title Case)

Version 0.0.0.9000

Description What the package does (one paragraph).

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`ds.cast_to_factor` *Cast specified columns to factor on a server-side data frame*

Description

Convert one or more columns of a server-side data frame to factor using the `cast_to_factorDS` server-side function. The resulting data frame is assigned as a new object on each server.

Server function called: `cast_to_factorDS`

Usage

```
ds.cast_to_factor(df, columns, modified_obj = NULL, datasources = NULL)
```

Arguments

df	A character string specifying the name of the server-side data frame containing the columns to be converted.
columns	A character vector specifying the names of the columns to convert to factor.
modified_obj	A character string specifying the name of the new server-side object to store the modified data frame. If NULL, a name is generated automatically by appending "_modified" to df.
datasources	A list of DSConnection-class objects obtained after login. If the datasources argument is not specified the default set of connections will be used: see datashield.connections_d

Value

Returns the name of the newly created server-side object containing the modified data frame.

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ds.check_missing_data *Filter columns of a server-side object based on amissingness threshold*

Description

Filters the columns of a server-side data set by removing variables whose proportion of missing values exceeds a user-defined threshold. The function can operate in two modes:

- "combined": computes a weighted global missingness across all servers and removes the same set of variables from every server.
- "split": applies the threshold independently on each server and removes server-specific sets of variables.

If no variables exceed the threshold, no filtering is applied and a message is returned indicating that no columns need to be removed.

Server function called: preprocess_dataDS, remove_columnDS

Usage

```
ds.check_missing_data(
  df,
  threshold,
  type = c("combined", "split"),
  preproc_newobj = NULL,
  datasources = NULL
)
```

Arguments

df	A character string providing the name of the input data frame object stored on each server.
threshold	A numeric value between 0 and 100 representing the maximum acceptable percentage of missing values per column.
type	A character string specifying the filtering strategy. Must be one of "combined" or "split".

`preproc_newobj` A character string specifying the name of the new pre-processed object to be created on each server. If `NULL`, a name is generated automatically by appending `"_preprocessed"` to `df`.

`datasources` a list of `DSConnection-class` objects obtained after login. If the `datasources` argument is not specified the default set of connections will be used: see `datashield.connections_`

Value

A character string giving the name of the newly created server-side object containing the pre-processed data set. If no columns are removed, the original data are left unchanged and the name of the intended output object is still returned invisibly.

`ds.check_patient_visit` *Check patient-visit combinations on a server-side data frame*

Description

Verify all patient-visit combinations in a server-side data frame and report any duplicates using the `check_patient_visitDS` server-side function. Messages are printed for each server indicating whether duplicates are present.

Server function called: `check_patient_visitDS`

Usage

```
ds.check_patient_visit(df, datasources = NULL)
```

Arguments

`df` A character string specifying the name of the server-side data frame containing at least the columns `pat_ID` and `Visit_months_from_diagnosis`.

`datasources` A list of `DSConnection-class` objects obtained after login. If the `datasources` argument is not specified, the default set of connections will be used: see `datashield.connections_`

Value

Invisible `NULL`. Messages are printed to indicate whether duplicate patient-visit pairs were found on each server.

ds.check_types *Check variable types on a server-side data frame*

Description

Verify that numeric and categorical variables in a server-side data frame comply with predefined valid ranges and values. Uses the server-side functions `check_numericDS` and `check_categoricalDS` to identify non-compliant columns. Messages are printed for each server indicating which columns, if any, are invalid.

Server functions called: `check_numericDS`, `check_categoricalDS`

Usage

```
ds.check_types(df, datasources = NULL)
```

Arguments

`df`	A character string specifying the name of the server-side data frame containing numeric and categorical variables to be checked.
`datasources`	A list of `DSConnection-class` objects obtained after login. If the `datasources` argument is not specified, the default set of connections will be used: see `datashield.connections_`

Value

Invisible `NULL`. Messages are printed to indicate which columns, if any, are non-compliant on each server.

ds.check_variables *Check variables on a server-side data frame*

Description

Verify that a server-side data frame contains exactly the expected variables. Uses the server-side function `check_variablesDS` to identify missing or extra variables. Messages are printed for each server indicating discrepancies.

Server function called: `check_variablesDS`

Usage

```
ds.check_variables(df, variables, datasources = NULL)
```

Arguments

`df`	A character string specifying the name of the server-side data frame to check.
`variables`	A character vector specifying the expected variable names.
`datasources`	A list of `DSConnection-class` objects obtained after login. If the `datasources` argument is not specified, the default set of connections will be used: see `datashield.connections_`

Value

Invisible list of results returned by the server-side function. Each element corresponds to a server and contains two components: `missing` (variables present in `variables` but missing in the data frame) and `extra` (variables present in the data frame but not in `variables`).

`ds.fill_missing_data` *Fill missing visit data on a server-side data frame*

Description

Invoke the server-side function `fill_missing_dataDS` to fill missing values in a specified column within each patient, ordered by visit time from diagnosis. The modified data frame is assigned to a new server-side object.

Server function called: `fill_missing_dataDS`

Usage

```
ds.fill_missing_data(
  df,
  pat_id_col,
  visit_col,
  value_col,
  filled_newobj = NULL,
  datasources = NULL
)
```

Arguments

`df`	A character string specifying the name of the server-side data frame containing at least the patient identifier column, the visit time column, and the variable to be forward-filled.
`pat_id_col`	A character string specifying the name of the patient identifier column.
`visit_col`	A character string specifying the name of the visit time column.
`value_col`	A character string specifying the name of the column whose missing values should be forward-filled.
`filled_newobj`	A character string specifying the name of the new server-side object that will store the modified data frame. If not provided, the default name is `paste0(df, "_filled")`.
`datasources`	A list of `DSConnection-class` objects obtained after login. If not specified, the default set of connections is used: see `datashield.connections_default`.

Value

Invisibly returns the name of the newly created server-side object. A message is printed indicating where the result has been saved.

ds.global_imputation *Global imputation of a server-side object using MICE*

Description

Performs imputation on data from a single source on the server-side.

Server function called: `global_imputationDS`

Usage

```
ds.global_imputation(
  df,
  id_col = NULL,
  m = 5,
  maxit = 30,
  imputed_newobj = NULL,
  datasources = NULL
)
```

Arguments

`df`	A character string specifying the name of the server-side data frame on which the global imputation is performed.
`id_col`	A character string indicating the name of the identifier column that is excluded from imputation.
`m`	A numeric value representing the number of imputations to generate. Default is 5.
`maxit`	A numeric value representing the maximum number of MICE iterations per imputation chain. Default is 30.
`imputed_newobj`	A character string specifying the name of the new imputed object to be created on each server. If NULL, a name is generated automatically by appending `"_imputed"` to `x`.
`datasources`	A list of DataSHIELD connections. If NULL, the active connections returned by `datashield.connections_find()` are used.

Value

The name of the newly created server-side object containing the imputed data set.

ds.remove_columns *Remove specified columns from a server-side data frame*

Description

Remove one or more columns from a server-side data frame using the `remove_columnsDS` server-side function. The resulting data frame is assigned as a new object on each server.

Server function called: `remove_columnsDS`

Usage

```
ds.remove_columns(df, col_names, newobj = NULL, datasources = NULL)
```

Arguments

`df`	A character string specifying the name of the server-side data frame from which columns will be removed.
`col_names`	A character vector specifying the names of the columns to remove. These columns must exist in the data frame, otherwise the server-side function will throw an error.
`newobj`	A character string specifying the name of the new server-side object to store the modified data frame. If NULL, a name is generated automatically by appending `"_filtered"` to `df`.
`datasources`	a list of `DSConnection-class` objects obtained after login. If the `datasources` argument is not specified the default set of connections will be used: see `datashield.connections_`

Value

Returns the name of the newly created server-side object containing the modified data frame.

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